

INDUSTRIAL PRODUCT DATASHEET

reComputer J601 Carrier Board for Jetson AGX Thor™ T5000 / T4000

reComputer J601 Carrier Board for Jetson AGX Thor™ T5000 / T4000: a 168mm x 153mm, up to 2070 TFLOPS embodied AI platform with 48V input, rich I/O, and up to 8x GMSL2 camera support for humanoid robotics, VLA, production AI, an...

SKU

100066562

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PRICE

\$579.00(Excl. VAT)



Overview

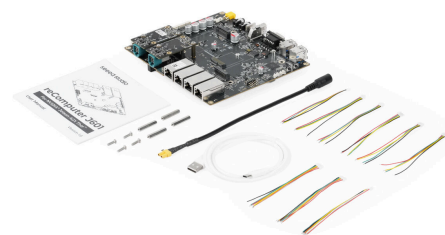
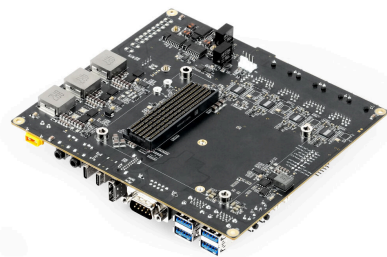
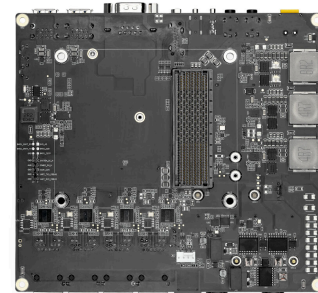
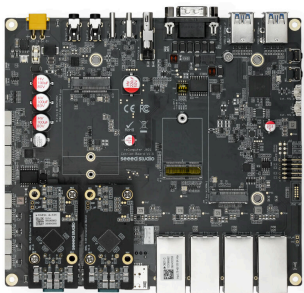


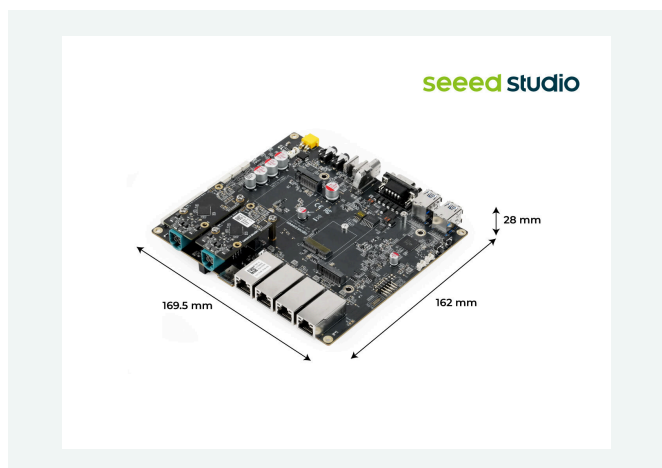
The reComputer J601 Carrier Board for Jetson AGX Thor™ T5000 / T4000 is a powerful embodied AI platform compatible with Jetson AGX Thor modules, offering up to 2070 TFLOPS in a 168mm x 153mm footprint with 48V battery input (max 54V), rich I/O including USB, 10G Ethernet, CAN, RS-485, and I2S, and vision AI support via 4x 4-lane CSI output and up to 8x GMSL2 cameras. It is designed for humanoid robotics, VLA development, production AI and automation, and agentic AI at the edge.

Key Features

- **Powerful embodied AI Platform** Compatible with the Jetson AGX Thor module, offering up to 2070 TFLOPS. Perfect edge platform for embodied AI development.
- **Compact Footprint** 168mm x 153mm footprint, suitable for robot prototyping and development, especially robots that has tight footprint such as humanoids
- **Wide Voltage Input Range** Can be used in 48V battery power system (max 54V input)
- **Rich IO capabilities** Includes most common IOs used in robotics prototyping, such as USB, 10G Ethernet, CAN, RS-485 and I2S
- **Vision AI Support** Features 4x 4-lane CSI output, and can be connected up to 8x GMSL2 cameras, making it ideal for vision AI applications such as BEV, Occupancy Grid, SLAM etc

Product Gallery





Specification

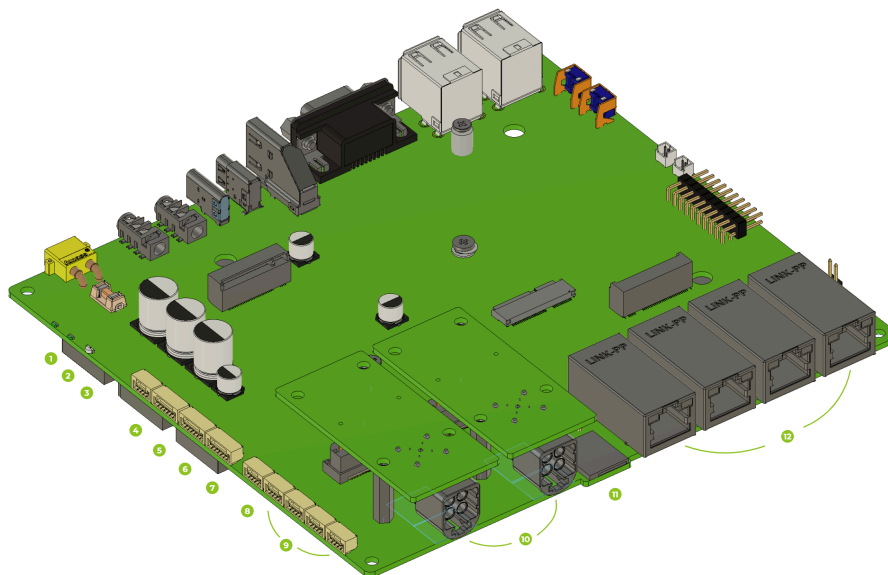
Product	reComputer J601 Carrier Board	
Storage	1x M.2 Key M slot	
Networking	1x M.2 Key E slot for WiFi/Bluetooth module 1x M.2 Key B slot for 4G/5G module	
I/O	USB	4x USB 3.2 Type-A (10Gbps); 1x USB 3.0 Type-C (Recovery); 1x USB 2.0 Type-C (Debug);
	RJ45	4x 10G Ethernet
	GMSL	8x GMSL2 (MAX96712 as deserializer)
	CAN	4x CAN (CAN-FD supported)
	RS485	1x RS-485
	RS232/422/485	1x RS-232/422/485
	I2S	1x I2S
	I2C	1x I2C
	Audio	1x MIC 1x Audio Output
	GPI/GPO	4x GPI 4x GPO
	Display	1x HDMI 2.1
	Fan	2x 4 pin Fan Connector (12V PWM)
	Button	1x Reset Button; 1x Recovery Button
	RTC	1x RTC 2-pin
Power	1 x XT30 Connector (19-48V DC)	
Mechanical	Dimensions (W x D)	168mm x 155mm
	Weight	312g
	Installation	Desk, wall-mounting
Operating Temperature	-10°C~60°C (with thermal grease) -10°C~55°C (with thermal pad)	
Warranty	2 Year	

Manufacturer	Model	Resolution
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Sensing	SG3S-ISC031C-GMSL2F	1920H*1536V
	SG2-AR0233C-5200-G2A	1920H*1080V
	SG2-IMX390C-5200-G2A	1920H*1080V
	SG8S-AR0820C-5300-G2A	3840H*2160V
Orbbec	Orbbec Gemini 335Lg	3D

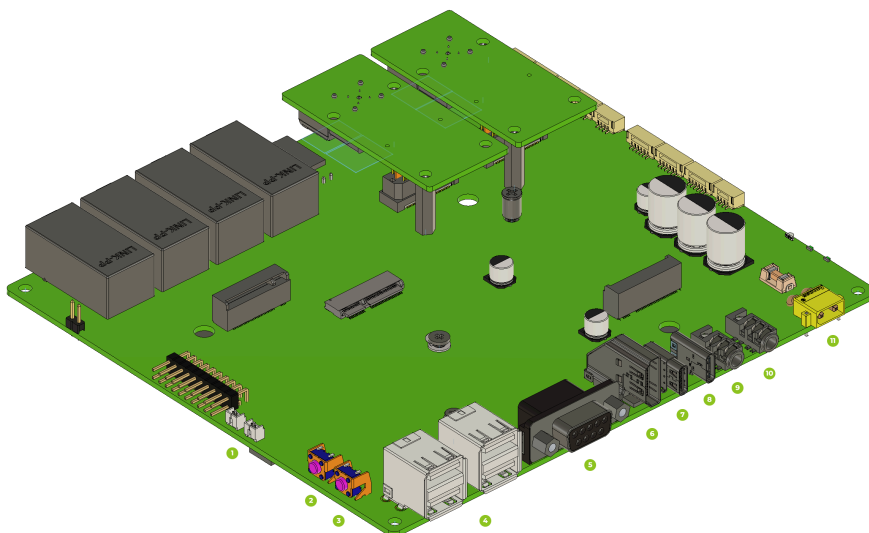
Hardware Overview

Side View 1



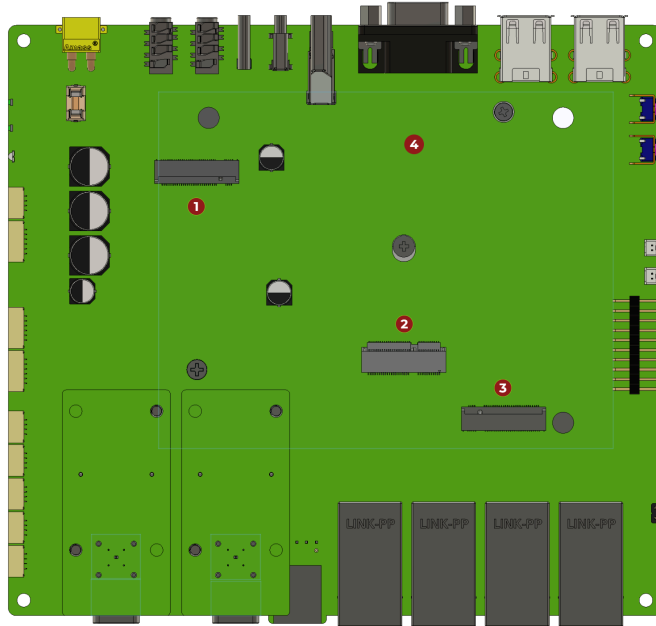
- 1 ACT LED x 1
- 2 Power LED x 1
- 3 SSD LED x 1
- 4 I2C x 1
- 5 I2S x 1
- 6 GPI x 4
- 7 GPO x 4
- 8 RS-485 x 1
- 9 CAN x 4
- 10 GMSL2 x 8
- 11 Nano SIM
- 12 10 GbE x 4

Side View 2



- 1 RTC 2-pin Connector
- 2 Recovery Button
- 3 Reset Button
- 4 USB 3.2 x 4
- 5 RS-232/422/485 x 1
- 6 Display
- 7 USB Type-C Recovery
- 8 USB Type-C Debug
- 9 Audio Input
- 10 Audio Output
- 11 DC 19~48 V

Bottom View



- 1 M.2 Key B Slot
- 2 M.2 Key E Slot
- 3 M.2 Key M Slot
- 4 RS-232/422/485 Mode Switch

Applications

- **Humanoid Robotics:** reComputer J601 offers all necessary I/O's for humanoid development, such as CAN, 10Gb Ethernet, RS485, Audio etc. Along with NVIDIA Jetson AGX Thor modules, reComputer J601 serves as the onboard state-of-the-art "brain" for these machines, enabling them to perceive, reason, plan, and act in real-time.
- **VLA development:** Jetson Thor is the edge platform purpose-built for physical AI, transforms VLA models from research curiosities into production-ready intelligence. It eliminates the cost, complexity, and latency of external GPU dependency. It enables robots that understand natural language, perceive complex environments, and act with human-like fluidity. And it does all of this at the edge—secure, responsive, and scalable. NVIDIA provides a seamless end-to-end workflow for VLA development on Jetson Thor. The Isaac GR00T foundation model family—including GR00T N1.5, N1.6, and N1.7—offers cutting-edge VLA capabilities that bridge perception and execution. With JetPack 7.x, CUDA 13, and TensorRT acceleration, developers can fine-tune models on cloud infrastructure and deploy them directly to Thor with minimal friction.
- **Production AI and Automation:** Jetson AGX Thor brings server-class AI inference directly to the factory floor. It enables robots that visually inspect products, understand verbal instructions, and coordinate with other machines simultaneously—all without offloading processing to the cloud. Equipped with a rich array of I/O interfaces—including 10Gb Ethernet, GMSL, and RS-485—the reComputer J601 serves simultaneously as a sensor hub and a real-time processing platform, enabling rapid deployment and production-ready integration.
- **Agentic AI on edge:** Jetson AGX Thor brings server-class reasoning to the factory floor, the warehouse, and the inspection line. It enables AI agents that don't just answer questions, but also take action. Paired with NVIDIA Nemoclaw, NVIDIA provides a complete, production-grade agentic AI software stack. So reComputer J601 can be your AI agent on the edge without burning tokens. It can monitor, decide, coordinate, and execute. And it does it all at the edge—secure, responsive, and scalable.

Part List

Seeed Carrier Board (reComputer J601)	x1
GMSL Extension Board (GMSL x4)	x2
XT30 to DC power cable	x1
USB Type-C to Type-A cable	x1
GH-1.25 4-pin 100mm JST Cable	x6
GH-1.25 6-pin 100mm JST Cable	x3
M3*25 Stud	x4

Compliance & Logistics



HSCODE	8473309000
USHSCODE	8473309100
UPC	
EUHSCODE	8473308000
COO	CHINA

FCC Requirement

Any changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

- (1) this device may not cause harmful interference, and
- (2) this device must accept any interference received, including interference that may cause undesired operation.

15.21

This transmitter must not be co-located or operating in conjunction with any other antenna or transmitter.

Note: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules.

These limits are designed to provide reasonable protection against harmful interference in a residential installation.

This equipment generates, uses, and can radiate radio frequency energy.

If not installed and used in accordance with the instructions, it may cause harmful interference to radio communications.

However, there is no guarantee that interference will not occur in a particular installation.

If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

Resource Links

- <https://www.seeedstudio.com/reComputer-J601-Carrier-Board-for-Jetson-AGX-Thor-p-6937.html>
- [Datasheet](#)
- [Carrier Board Schematic](#)
- [3D File](#)
- [Seeed Nvidia Jetson Product Catalog](#)
- [Seeed Nvidia Jetson Success Cases](#)
- [Seeed Jetson AGX One Pager](#)

Seeed Technology Co.,Ltd.

9F, Building G3, International E City,
Zhongshanyuan Road, Nanshan, Shenzhen, China

Tel: +86 0755-80695676

Web: www.seeed.cc

Shop: www.seeedstudio.com